

## Lab Exercise #3: Calculating a What If Semester Average

### 1 Overview

- Write a C program to determine the final exam score you need to earn your desired semester average.
- Use your syllabus to identify the components and percentages needed in the calculation. The percentages must be global constants (`#define`) at the top of your program.
- Request user input for each component of the semester average.
- Calculate the required final exam score and print it to the screen.

### 2 Code Documentation

The very first lines in *all* of your lab project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, for the lab exercises, should be the name of the assignment, your name and your lab partners name, e.g:

```
printf("Lab #3 - Joe Student and Jane Doe\n");
```

The next line of output should be `printf("EGRE 245: Introduction to Programming Using C, September 16\n");`

Please fill in the correct date.

### 3 Sample Run

Your output should match the sample run below.

```
Lab 3: Mark Schwitzerlett and Robert Redford  
September 16, 2025
```

```
What average do you want to earn: 79.5
```

```
Input your Reading Assignment average: 92
```

```
Input your Homework average: 85
```

```
Input your Lab Assignment average: 88
```

```
Input your Programming Assignment average: 83
```

```
Input your Quiz average: 78
```

```
Input your TopHat average: 95
```

```
Input your Midterm Score: 72
```

```
You need a 64.3 on the final exam to earn a 79.5 in the course
```

```
...Program finished with exit code 0
```

```
Press ENTER to exit console. 
```

## 4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Please name your file `Lab3_FirstName_LastName.c`.

**Due date:** The lab is due at the end of your lab period, either Tuesday, September 16, or Thursday, September 18th.